

INSTRUCTIONS TO CANDIDATES (Please read these instructions carefully)

- You have been given this question booklet and a separate answer sheet. The question booklet contains 50 questions
- Do any necessary rough work in this booklet.

SECTION A [20 MARKS]

- During a national census, the population of a country was 73, 842, 029. Determine the place value of digit 3 in the number.
 - A. Millions.
 - B. Hundreds of thousands.
 - C. Tens of millions.
 - D. Tens of thousands.
- James recorded the following numbers in a field book.
 $36\frac{3}{4}$, -9.7, 14.50, -2
 Which one of the following numbers is an integer?
 - A. -9.7.
 - B. 14.50.
 - C. $36\frac{3}{4}$.
 - D. -2.
- A floor of a room is squared in shape. the floor is covered with slabs. Each slab measures 60cm and 42cm. What is the least possible length of the floor given that the number of slabs used along the length and the width were fitting exactly?
 - A. 408cm.
 - B. 102cm.
 - C. 420cm.
 - D. 2520cm.
- The arrangement of four chairs P, Q, R, S, in a room was represented using the matrix below.

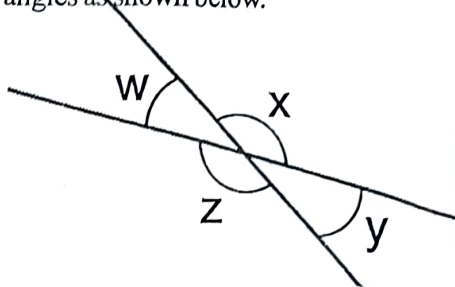
$$\begin{bmatrix} P & Q \\ R & S \end{bmatrix}$$

Which one of the following statements describes the position of R in the matrix.

- First row, second column.
- Second row, first column.
- First column, second row.
- Second column, first row.
- The scale used in a map is 1:50,000. A piece of land on the map is represented by a

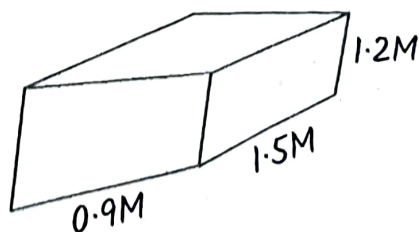
rectangle measuring 4cm and 2.5cm. What is the actual size of the land in hectares?

- A.25.
 - B.50.
 - C.250.
 - D.125.
6. A welder joined two steel rods to form four angles as shown below.



Which one of the following statements is true about the angles.

- A. $Y=180-X$.
 - B. $Y=360-X$.
 - C. $Y=180-W$.
 - D. $Y=360-(W+Z)$.
7. The figure below represents Muluma's water tank.



The tank has water to the height of 0.8m. How many litres of water are need to fill the tank completely?

- A.1080 Litres.
 - B.1620 Litres.
 - C.540 Litres.
 - D.2700 Litres.
8. Owiti sold a laptop for sh.34,000. He made a loss of 15%, which one of the following calculations gives the price at which he had bought the laptop.

- A. $34,000 \times 1.15$.
- C. $34,000 \div 0.85$.

8. $34,000 \times 0.85$. D. $34,000 \div 1.15$.
9. A number K is represented by an expression.

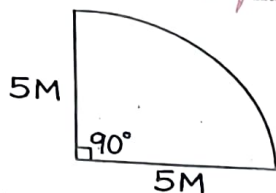
$$K = \frac{3a^2 + C}{bc}$$

$$K = \frac{3a^2 + c}{bc}$$

Given that $a=3$, $b=2$ and $C=a+b$

Calculate the value of K.

- A. -41. B. -14. B
C. -9.5. D. 14
10. A flower bed is in the shape as drawn below



What is the distance round the flower bed?

- A. 19.638m. C. 7.855m
B. 17.855m. D. 12.855m B
11. Salome deposited sh.45,000 in financial institution that paid a simple interest rate of 412% p.a. Which one of the following calculations will give the amount of money at the end of 3 years.

- A. $45,000 + 1350$. C. $45,000 + 4050$.
B. $45,000 + 2050$. D. $45,000 + 6075$. D

12. Two friends Isaiah and Mesh worked out $3^6 \div 3^2$.

Using the laws of indices which one of the following is the correct simplified form of their working.

- A. 3^{12} B. 3^8
C. 3^4 D. 3^3

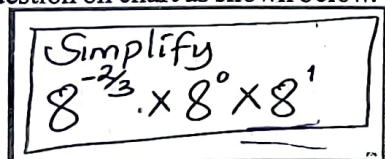
13. A builder needed six blocks for the first step of the staircase, ten blocks for the second step and fourteen blocks for the third step. If the pattern continues how many blocks will be needed for the seventh step.

- A. 22. B. 30 B
C. 34. D. 42.

14. Maya drew a regular polygon in her book. Each interior angle of the polygon measured 60° . How many sides did the polygon have.

- A. 6. B. 4. D
C. 5. D. 3.

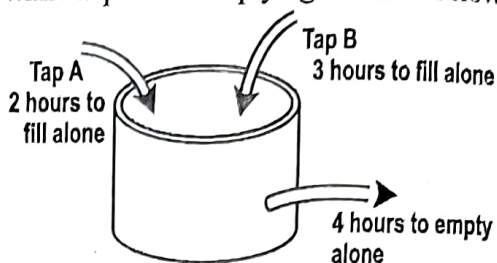
15. A grade eight teacher wrote the following question on chart as shown below.



The teacher asked the learners to solve the equation. Which one of the following is the correct solution of the question on the chart.

- A. $\frac{1}{2}$. B. 2.
C. 36. D. 16.

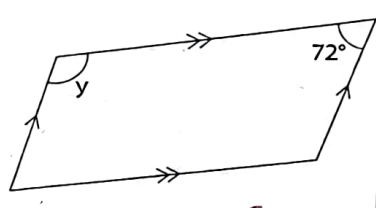
16. Tap A and Tap B below were filling the tank while Tap C was emptying as shown below.



Supposed the three taps are turned on simultaneously when tank is empty.

Determine how long it will take the two taps to fill the tank.

- A. 127 hours. B. 7 12 hours.
C. 115 hours. D. 157 hours.
17. Rony came up with a display stand in the shape of a parallelogram as shown below.



What is the size of the supplementary angle of Y.

- A. 108° . B. 72°
C. 90° D. 18°

18. Arianna wrote a reciprocal of a number as $\frac{2}{15}$.

What was the value of the original number.

- A. 511 B. 115
C. 125 D. 512

19. Frank recorded the speed of light in a vacuum as 300,000,000 m/s. What is this speed written in standard form.

- A. 3×10^8 B. 30×10^7
C. 300×10^6 D. 3000×10^5

20. Kilenzo recorded the temperature of a liquid as -43°K . Which one of the statements below gives the correct temperature in degrees Celsius?

- A. $-43 \div 273$. B. -43×273 .
C. $-43 - 273$. D. $-43 + 273$. C

SECTION B [80 MARKS]

21. Stacy a grade nine learner arranged all prime numbers less than ten in a descending order to form a number.

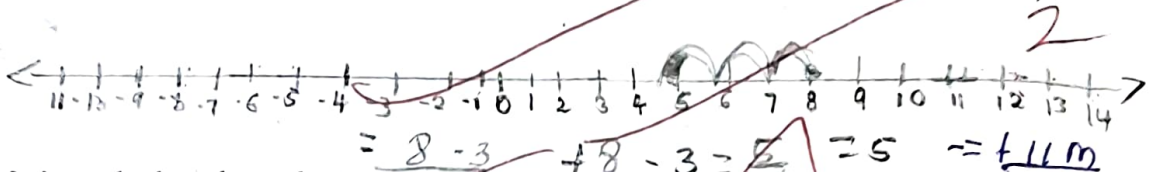
a). Write down the number formed. (2mks)

57 63 43 38 37 33 31 23 17

b). Determine the total value of the second digit in the number formed. (2mks)

57 63 43 38 37 33 31 23 17
 $57 = 50 + 7$
 $3 - 3 = 0$
 $7632 = 500$

22. A Crane operator working 8 metres above the ground level remembered that he left his tools 3 metres below the ground, using a number line, determine the total distance the operator will cover to retrieve his tool box to continue with his work at his working station. (4mks)



23. A fruit vendor bought apples for her business. She sold $\frac{1}{5}$ of the apples on the first day, $\frac{1}{4}$ of the apples on the second day and 23 of the remaining apples on the third day. Determine the number of apples the vendor had at the beginning of the first day. (5mks)

Step 1: Sold $\frac{1}{5}$ first day
 Step 2: Sold $\frac{1}{4}$ second day
 Step 3: $\frac{1}{5} + \frac{1}{4} = \frac{4+5}{20} = \frac{9}{20}$
 Step 4: $\frac{9}{20} \times 23 = \frac{207}{20} = 10.35$
 Ans: 10.35

24. A company produced 15,625 Royco cubes, each side of the Royco cube had a length of 1.5cm. The Royco cubes were packed into a single large box. Given that all the cubes fitted exactly into the box, determine the length of one side of the box. (4mks)

Step 1: $15625 \div 15625 = 1$
 Step 2: $15625 \div 15625 = 1$
 Step 3: $15625 \div 15625 = 1$
 Step 4: $15625 \div 15625 = 1$
 Ans: 1

25. A farmer needed 12 workers each working 6 hours per day to harvest maize in his farm for 5 days. Determine how many more workers the farmer will require to complete the same task in 3 days if each works for 8 hours per day. (4mks)

Step 1: Workers hours days
 12 6 5
 Step 2: $12 \times 6 \times 5 = 360$
 Step 3: $360 \div 8 \div 3 = 15$
 Step 4: $15 - 12 = 3$
 Ans: 3

26. Lupita bought some oranges, Joe bought 4 more oranges than Lupita, they both bought a total of 18 oranges. Determine the number of oranges Joe bought. (3mks)

Step 1: Joe bought 4
 Step 2: $x + 4 = 18 - 4 = 14$
 Step 3: $18 - 4 = 14$
 Step 4: $14 \div 2 = 7$
 Ans: 7

27. The sum digits of a two-digit number is 8, when the digits are interchanged, a new number formed becomes 54 less than the original number. Determine the original number. (5mks)

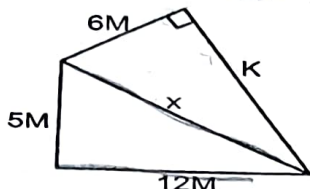
Step 1: Sum $\rightarrow 18$
 Step 2: $54 \div 9 = 6$
 Step 3: $x + y = 8$
 Step 4: $x + y = 8$
 Ans: 13

28. Okeri and Tumbo estimated their heights. Okeri indicated that his height was greater than 1.6m, while Tumbo indicated his height was not more than 1.9m. Write a compound inequality statement to represent the heights of Okeri and Tumbo. (2mks)

Okeri greater $> 1.6m$
Tumbo $\leq 1.9m$

$$1.6 < 1.9$$

29. A flower garden is in the shape shown below.



- a). Calculate the length of the side marked X and K. (3mks)

Step 1
 $h^2 = b^2 + c^2$
 $5^2 = 6^2 + x^2$
 $25 = 36 + x^2$
 $x^2 = 25 - 36$
 $x^2 = -11$

Step 2
 $6^2 + 5^2 = 13^2$
 $36 + 25 = 169$
 $61 = 169$

Step 3
 $\frac{1}{2} \times b \times h$
 $\frac{1}{2} \times 11 \times 6$
 33

Step 4
 $\frac{1}{2} \times b \times h$
 $\frac{1}{2} \times 3.31 \times 6$
 9.93

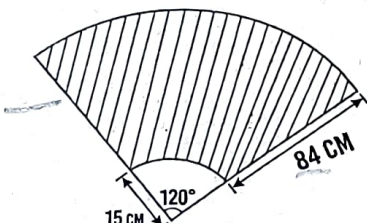
- b). Use a mathematical table to find the area of a square garden that may be formed at the side marked K. (2mks)

Step 1
 $K = 9.93$

Step 2
 $9.93^2 = 98.6049$

Step 3
 3.14×12
 37.68

30. Maria moved the car windscreen wiper and observed the area swept by the wiper blade. She represented her observation as shown below in the diagram. The shaded part represents the area swept by the blade.



Determine the area of the shaded part (5mks)

(Take $\pi = 3.142$)

Step 1
 $\frac{120}{360} \times 3.142 \times 15 \times 84$
 $= 1319.64$

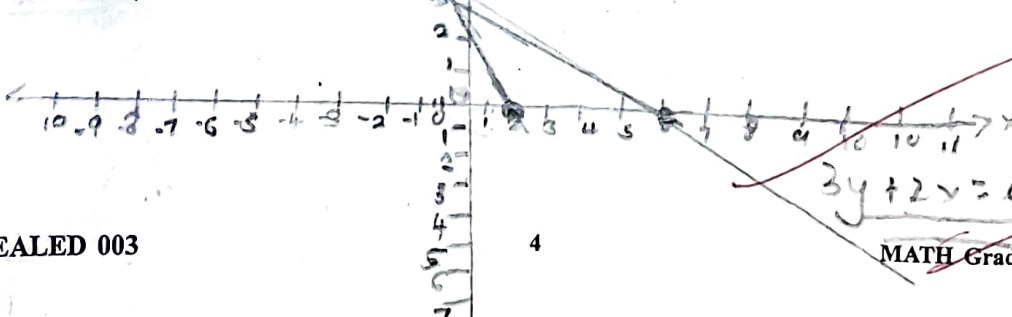
Step 2
 $\frac{120}{360} \times 3.142 \times 15 \times 84$
 $= 235.65$

Step 3
 $\frac{120}{360} \times 3.142 \times 42 \times 42$
 $1847.496 + 235.65$
 $= 2083.146 \text{ cm}^2$

31. The line drawn below has an equation of $3y + 2x = 6$.

A $3y + 2x = 6$ B

The line crosses the x-axis and y-axis at points S and T respectively. Determine the co-ordinates of S and T. (4mks)



32. A school tank has a water tank in the shape of a cylinder; the diameter of the tank is 2.4m and the height is 1.8m. the tank is filled with 2036 litres of water. Determine the fraction of the tank that is filled with water. (4mks)



1.8

$$\frac{22}{7} \times 1.2 \times 1.2 \times 1.8 = 4625714286$$

$$= \frac{22}{7} \times 2.4$$

$$\frac{22}{7} \times 2.4 \times 1.8$$

$$\frac{22}{7} \times 2036$$

$$\begin{array}{r} 44792 \\ 7 \\ \hline 6393 \end{array}$$

33. Odhiambo recorded the mass of a fish as 4.218kg. represent the mass of the fish as a fraction. (4mks)

Step 1

$$1 \text{ kg} \rightarrow 1000 \text{ gram}$$

$$4.218 \text{ kg} \rightarrow ?$$

Step 2

$$4.218 \text{ kg} = \frac{4218}{1000}$$

Step 3

$$\frac{4218}{1000}$$

34. Makori keeps chickens, ducks and turkeys in his farm. The ratio of chicken to ducks is 4:3 and ducks to turkeys is 5:2, Determine the ratio of chicken to turkeys in the simplest form. (4mks)

$$\frac{4}{3} \times \frac{5}{2} = \frac{20}{6} = \frac{10}{3}$$

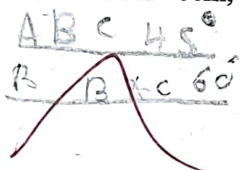
$$\frac{4}{3} \times \frac{5}{2} = \frac{20}{6} = \frac{10}{3}$$

$$\frac{10}{3} = \frac{10}{3}$$

$$\frac{10}{3} = \frac{10}{3}$$

35. The cross section of a roof is in the shape of a triangle. In a drawing, the cross section is represented using triangle ABC such that AB=6cm, $\angle BAC = 25^\circ$ and $\angle BAC = 35^\circ$

a) Draw triangle ABC. (3mks)



b). Drop a perpendicular line from C to meet AB at N. Measure CN. (2mks)

36. The cash price of a motorbike is sh.120,000. Okitoi bought the motorbike on hire purchase terms, He paid a deposit of sh.15,000 and paid the balance in twelve equal monthly instalments of 12,000 each. How much more was the hire purchase price more than the cash price. (4mks)

$$CP = D + M + T$$

$$120,000 = 15,000 + (12,000 \times 12) + T$$

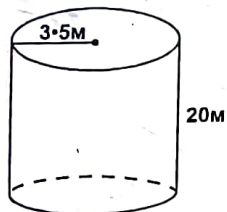
$$120,000 = 15,000 + 144,000 + T$$

$$120,000 = 159,000 + T$$

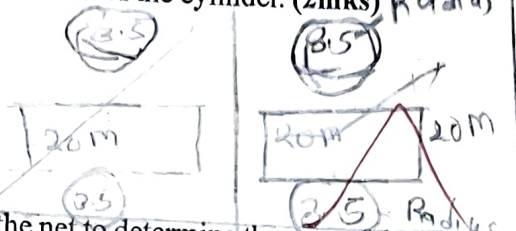
$$T = 120,000 - 159,000$$

$$T = -39,000$$

37. A sewage company designed the cylindrical pipe show below for transportation of waste from point A to B during a certain process in the factory. The pipe has a radius of 3.5m and a height of 20m as shown below.



a). Draw the net of the cylinder. (2mks) Radius



b). Use the net to determine the area of the material required to make the cylindrical shape. (2mks)

$$\frac{22}{7} \times 3.5 \times 3.5 = 38.5$$

$$\frac{22}{7} \times 3.5 \times 3.5 = 38.5$$

$$\frac{22}{7} \times 3.5 \times 20 = 2200$$

$$38.5 + 38.5 + 2200 = 2277$$

38. Rehema recorded the sizes of shoes worn by some learners in her class as follows: 34, 35, 38, 36, 38, 37, 36, 38, 39, 38

a). Prepare a frequency distribution table for the information given above. (2mks)

b). Determine the.

(i). Medium. (1mk)

$$34 + 37 = 71 \quad \frac{71}{2} = 35.5$$

(ii). Mean. (2mks)

$$34 + 35 + 36 + 38 + 37 + 36$$

$$34 + 35 + 36 + 38 + 37 + 36 + 38 + 39 + 38$$

$$= 333$$

39. Saruk represented the number of bags of beans, wheat, maize and peas harvested using a matrix below.

$$\begin{bmatrix} 5 & 9 \\ 13 & -7 \end{bmatrix} \quad \begin{bmatrix} 8 & 11 \\ 6 & 4 \end{bmatrix}$$

Matrix A represents the harvest in the first season while B harvest of the second season that year. Find the value of A+B, the total harvest in that year. (2mks)

$$\begin{bmatrix} 5 & 9 \\ 13 & -7 \end{bmatrix} + \begin{bmatrix} 8 & 11 \\ 6 & 4 \end{bmatrix} = \begin{bmatrix} 13 & 20 \\ 19 & -3 \end{bmatrix}$$

40. Amanda has a packet containing 3 red pens, 2 black pens, and 4 blue pens. She picked a pen at random from the packet. Determine the probability that the pen she picked was:

a). Red. (2mks)

$$\frac{3}{9} \times 4 = \frac{4}{3}$$

$$\frac{3}{9} \times 10 = \frac{10}{3}$$

$$\frac{3}{9} = \frac{1}{3}$$

$$= \frac{1}{3}$$

b). Blue. (1mk)

$$\frac{4}{9} \times \frac{2}{3} = \frac{8}{27}$$

$$\frac{3}{9} \times \frac{2}{3} \times 4 = \frac{8}{9}$$

$$\frac{2}{9} \times 7 = \frac{14}{9}$$

$$\frac{8}{9} + \frac{14}{9} = \frac{22}{9}$$

$$\frac{4}{9} \text{ probability } \frac{4}{9} = \frac{4}{9}$$